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(54) **MONATOMIC ELECTROLYTE AND
ELECTROCHEMICAL PROCESS
INCLUDING THE SAME**

(52) **U.S. Cl.**

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(71) Applicant: **Kyle Russell Dunn**, Aurora, CO (US)

(57)

ABSTRACT

(72) Inventor: **Kyle Russell Dunn**, Aurora, CO (US)

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A device and process that broadens the commercial and industrial applicability of metal extraction from ore by utilizing a novel substance as an electrolyte. The process overcomes the technical limitations of conventional electrolytic processes. A monatomic substance is used to create an electrolyte that avoids degradation, improves process kinetics, and minimizes end-product contamination. The electrolyte enables an electrochemical process at a lower operating temperature, offers a wider electrochemical potential window, and runs at higher reaction rates than either molten or aqueous processes. The device comprises a reactor, an electrochemical cell, a means for generating electromagnetic radiation, and a waveguide. The electrochemical cell is located within the reactor, and the means for generating electromagnetic radiation is coupled to the waveguide, and the waveguide is communicatively coupled to the reactor.

Related U.S. Application Data

(60) Provisional application No. 63/563,045, filed on Mar. 8, 2024.

Publication Classification

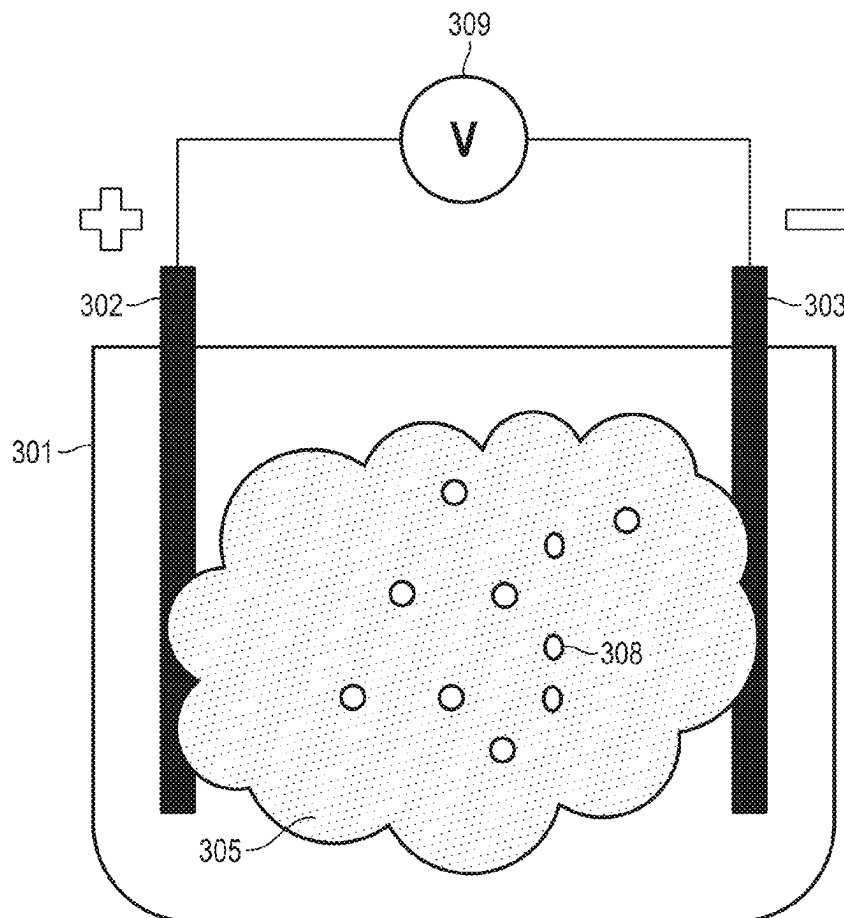
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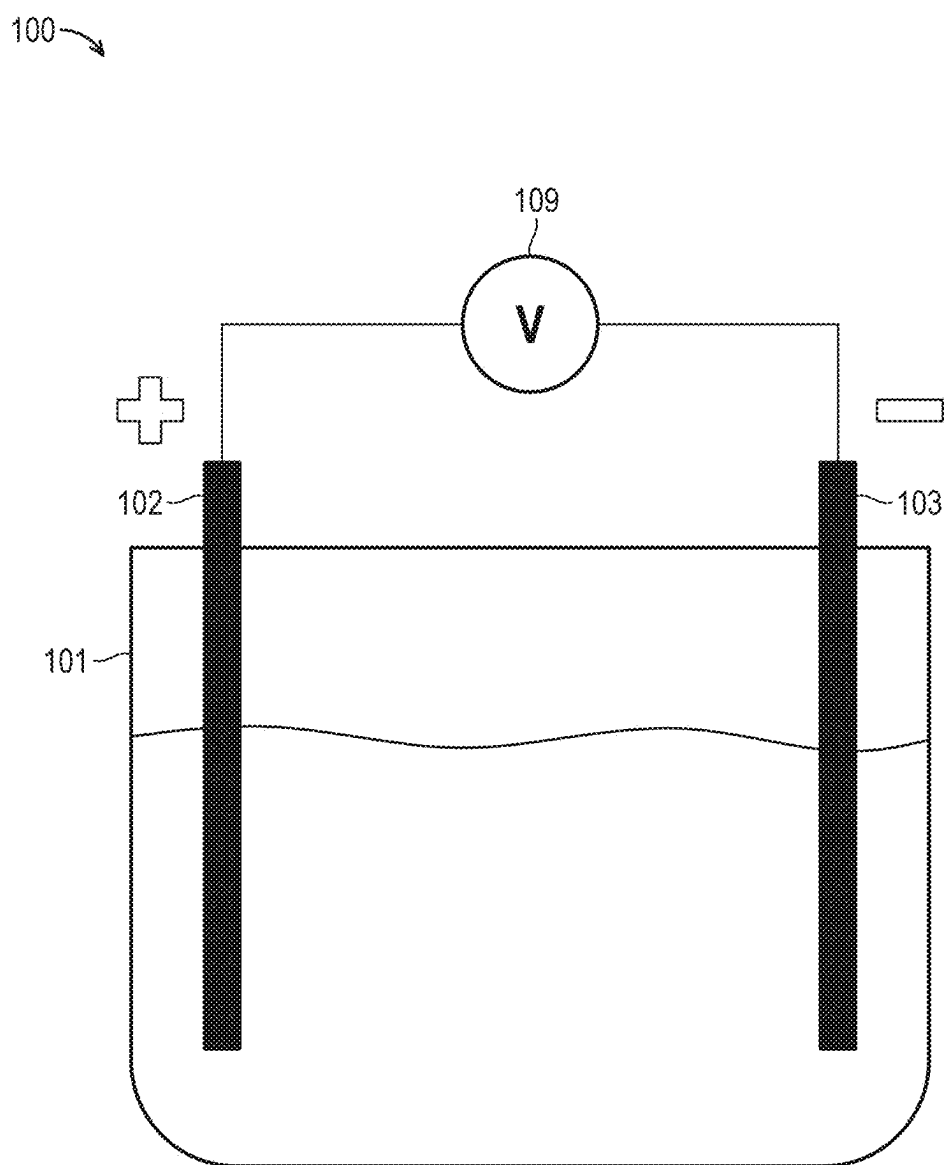


FIG. 1

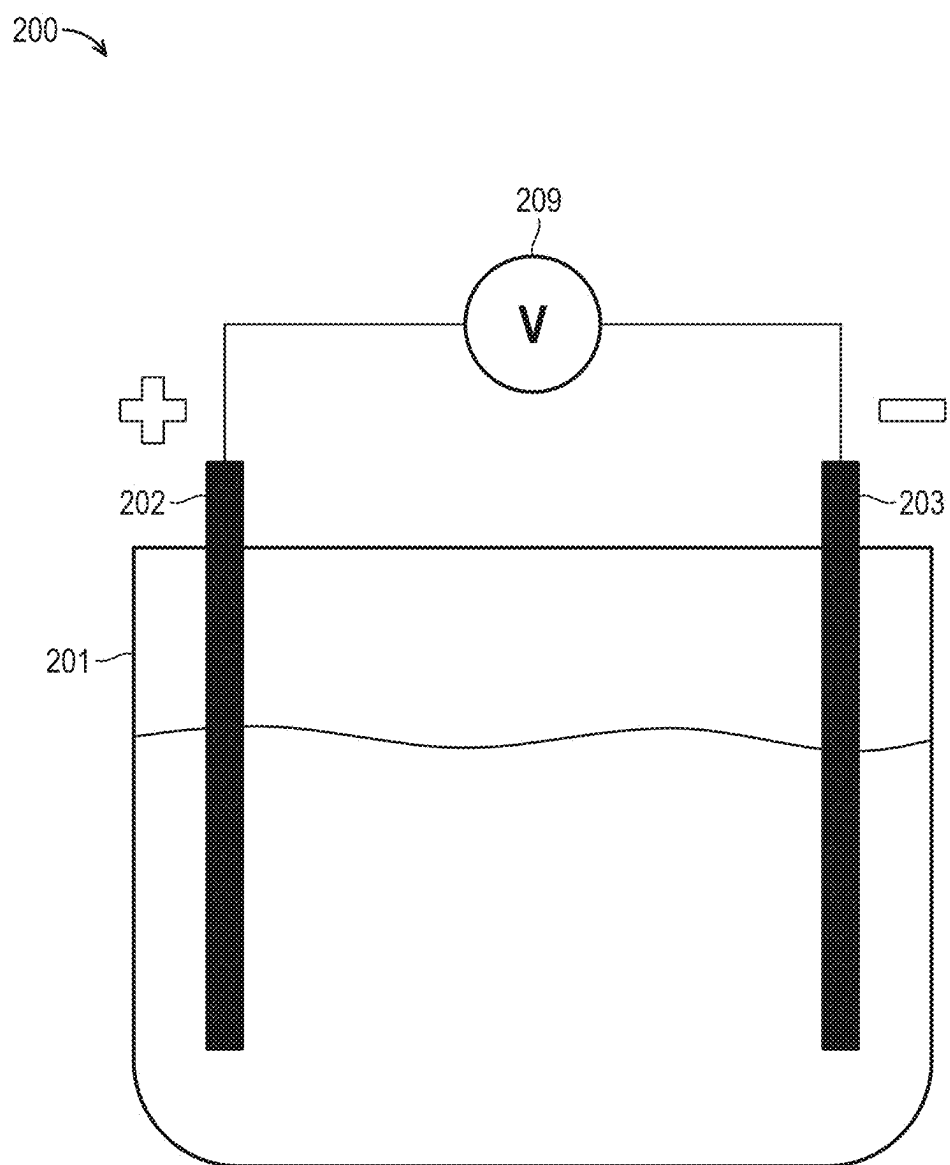


FIG. 2

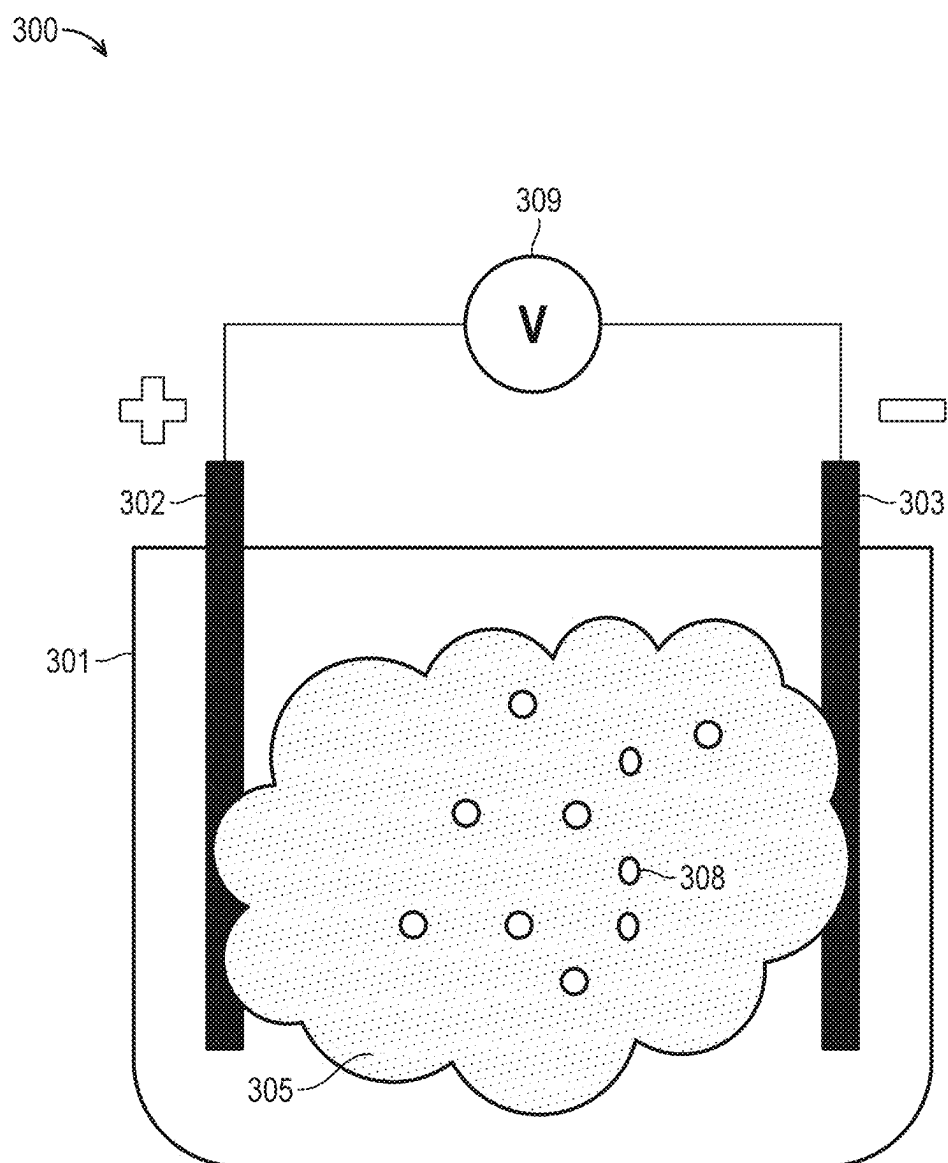


FIG. 3

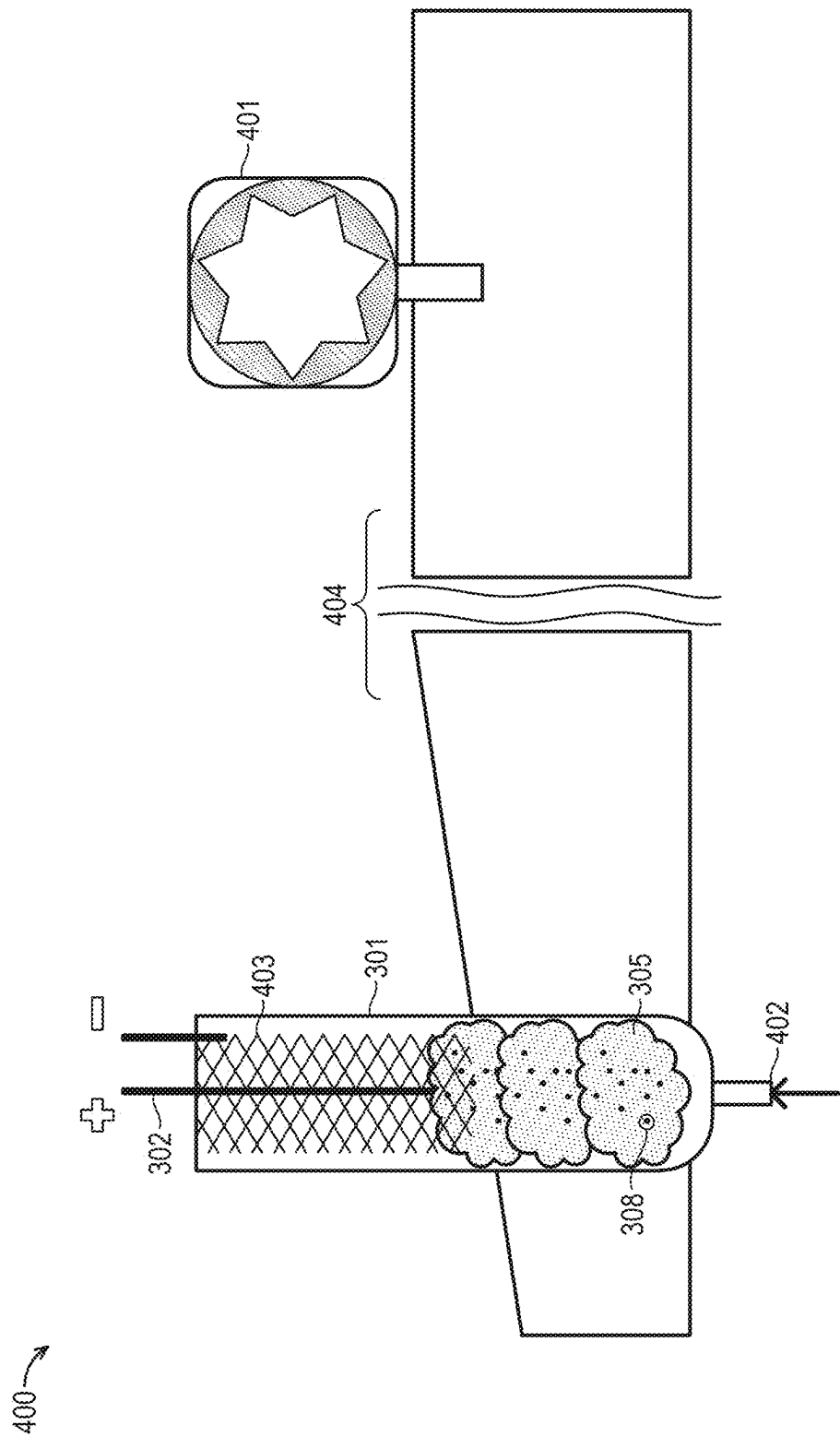


FIG. 4

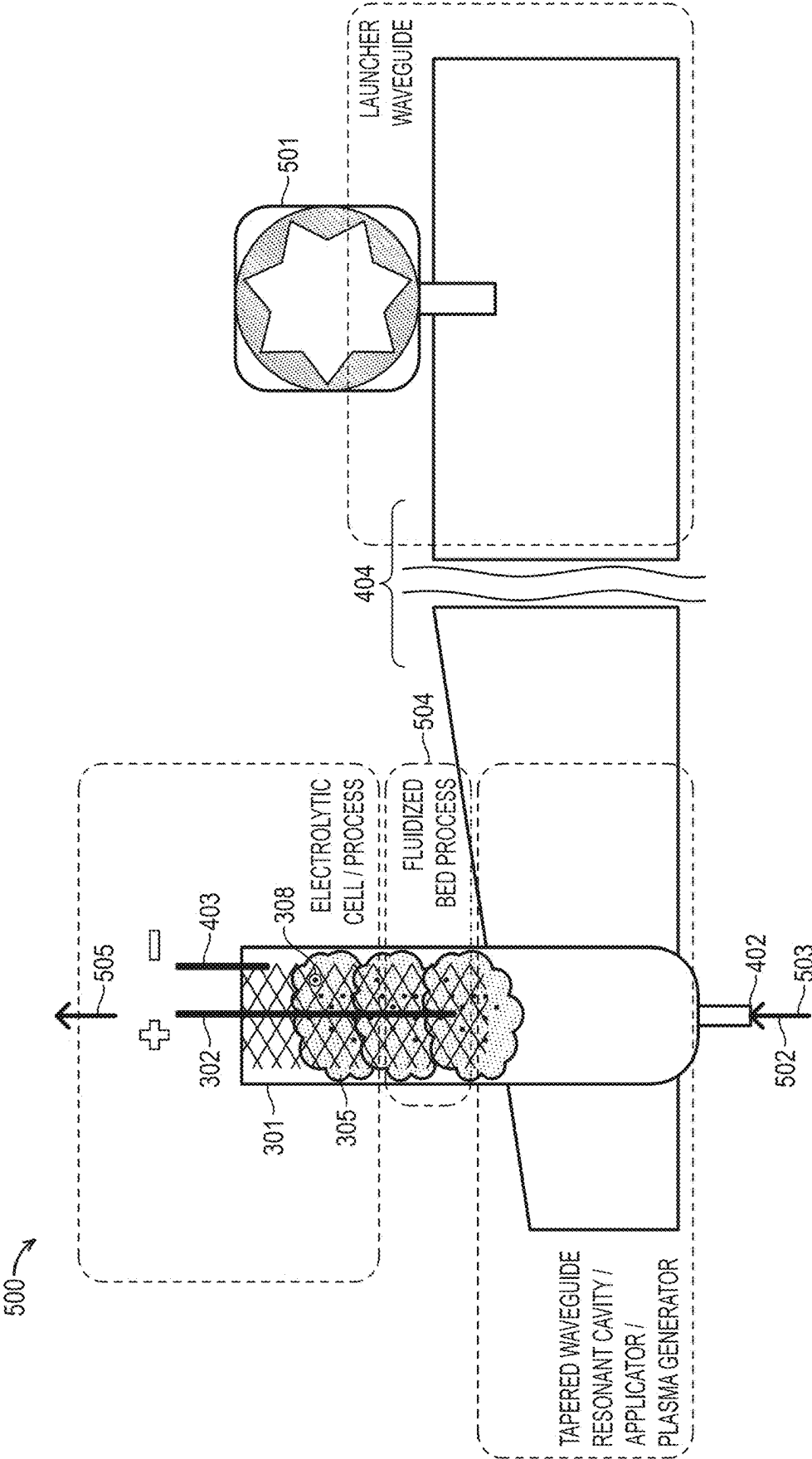


FIG. 5

MONATOMIC ELECTROLYTE AND ELECTROCHEMICAL PROCESS INCLUDING THE SAME

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to the co-pending U.S. application, Ser. No. 63/563,045, filed on Mar. 8, 2024, which is hereby incorporated by reference for all purposes.

BACKGROUND

[0002] The present disclosure relates generally to solid-state electroextraction. It describes a device and process for the electrolytic extraction of metals from ore. The present invention broadens the commercial and industrial applicability of electroextraction by utilizing a novel substance for the electrolyte to overcome the technical limitations of conventional electrolytic processes.

[0003] Electrolysis is the process by which an electric current is passed through a substance to effect a chemical change. A chemical change is one in which the substance loses (oxidation) or gains (reduction) an electron. The process is carried out in an electrolytic cell, an apparatus consisting of two electrodes (an anode and a cathode) apart and dipped into an electrolyte solution containing positively and negatively charged ions. The substance to be transformed may form the electrode, constitute a solution, or be dissolved in the solution. An electric current is applied across the anode and cathode. At the anode, oxidation occurs (loss of electrons), and reduction occurs (gain of electrons) at the cathode. FIG. 1 provides an example of a typical electrolytic cell used to split water into hydrogen and oxygen.

[0004] Electrolysis is used extensively in metallurgical processes, such as in extracting (electrowinning) or purifying (electrorefining) metals from ores. One method is molten salt electrolysis, which is a proven technology for the extraction of metals; all primary aluminum is produced in this manner. FIG. 2 depicts an example of this process.

[0005] The invention uses electrolysis to extract metals from a metal oxide or metal sulfide ore (collectively, feedstock). A monatomic substance, in this case a gas, is stimulated with electromagnetic radiation to form the electrolyte comprising ionized gas. Non-ionized and ionized gas is used as a carrier for the feedstock. An electric potential is applied across the electrodes, the metal is collected at the cathode, and the non-metal is evolved at the anode and exits the reactor with the gas.

[0006] Current methods have slow process kinetics, higher end-product contamination, and are limited by competing reactions to only certain ores. The traditional molten electrolyte processes are run at an elevated temperature ($>700^{\circ}\text{C}$). Both molten and aqueous electrolytes are limited to a narrow electrical potential window and exhibit slow process kinetics. Thus, there is a need for an improved electroextraction device and process that improves and advances the design of known electroextraction devices and methods. Examples of new and useful electroextraction devices and methods relevant to the needs of the field are discussed below.

SUMMARY

[0007] The invention is a device and process for the electrolytic extraction of metals from ore. It broadens the commercial and industrial applicability of electroextraction by utilizing a novel substance for the electrolyte to overcome the technical limitations of conventional electrolytic processes. The invention's use of a gaseous medium improves mass transport (i.e., process kinetics); the monatomic composition avoids degradation and minimizes end-product contamination.

[0008] The electrolyte enables electrochemical processes with a lower operating temperature ($<700^{\circ}\text{C}$.) compared to traditional molten electrolytes, a wider electrical potential window (0 to approximately 20V instead of 0 to 3.5V) compared to aqueous electrolytes, and better mass transport properties than either molten or aqueous electrolytes. The electroextraction device comprises a reactor, an electrochemical cell, a means for generating electromagnetic radiation, and a waveguide. The electrochemical cell is located within the reactor, and the means for generating electromagnetic radiation is coupled to the waveguide, and the waveguide is communicatively coupled to the reactor.

[0009] The device extracts metals from a feedstock by stimulating a medium comprising a monatomic substance with electromagnetic radiation until it is ionized. The ionized medium is electrically conductive and functions as an electrolyte. When the medium achieves sufficient electrical conductivity, an electrolytic process occurs within the reaction chamber.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 depicts a conventional aqueous electrolytic cell.

[0011] FIG. 2 depicts a conventional molten salt electrolytic cell.

[0012] FIG. 3 depicts the reactor and the electrolytic cell.

[0013] FIG. 4 depicts the reactor, electrolytic cell, and the electromagnetic excitation device.

[0014] FIG. 5 depicts a fluidized bed process.

DETAILED DESCRIPTION

[0015] The disclosed invention will be better understood by reviewing the detailed description and figures. The detailed description and figures provide merely examples of the various inventions described herein. Those skilled in the art will understand that the disclosed examples may be varied, modified, and altered without departing from the scope of the inventions described herein. Many variations are contemplated for different applications and design considerations; however, for the sake of brevity, each and every contemplated variation is not individually described in the following detailed description.

[0016] Throughout the following detailed description, examples of various embodiments are provided. Related features in the examples may be identical, similar, or dissimilar in different examples. For the sake of brevity, related features will not be redundantly explained in each example. Instead, the use of related feature names will cue the reader that the feature with a related feature name may be similar to the related feature in an example explained previously. Features specific to a given example will be described in that particular example. The reader should understand that a

given feature need not be the same or similar to the specific portrayal of a related feature in any given figure or example.

[0017] The invention extracts metals from a feedstock by stimulating a medium of a monatomic substance with electromagnetic radiation until said medium is ionized. The ionized medium is electrically conductive and functions as an electrolyte. When the medium achieves sufficient electrical conductivity, an electrolytic process occurs within the reaction chamber containing the electrolyte, an electrolytic cell, and feedstock.

[0018] The invention comprises a reactor. Within the reactor is an electrolytic cell comprising an anode and a cathode. The monatomic substance is an inert gas, also known as a noble gas, with non-limiting examples including Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), and Radon (Rn), with Argon being most preferred. The feedstock is a metal ore in either a metal oxide or sulfide. The feedstock is finely ground. Radiofrequency, microwave, or higher-frequency electromagnetic radiation can be used to ionize the gas. Non-limiting examples of electromagnetic radiation also include high-energy electromagnetic radiation, like X-rays or gamma rays.

[0019] Microwave radiation generated by a magnetron, particularly a cavity magnetron operating within an Industrial, Scientific, and Medical (ISM) band, is preferred. A solid-state transistor-based excitation device may also be used. The electromagnetic radiation in the 1 MHz to 1 THz range is directed to the reactor via a waveguide. A hollow tube waveguide with a rectangular, circular, or oval cross-section is preferred. The ionized gas envelopes the feedstock, bombarding it with free electrons, leading to dissociation into non-metal anions and metal cations. The conductive ionized gas facilitates electrical current flow between the anode and cathode, enabling mass transport for feedstock ions. The cations are reduced at the cathode, thereby extracting the desired metal, and the non-metal anions are oxidized at the anode: the non-metal and the gas exit through the top of the reactor.

Monatomic Electrolyte and Electrochemical Process Including the Same

[0020] With reference to FIGS. 3-5, MONATOMIC ELECTROLYTE AND ELECTROCHEMICAL PROCESS INCLUDING THE SAME will now be described. The monatomic electrolytes and processes discussed herein function to extract metals from ore via an electro-reduction process. An electrolyte is created by ionizing a noble gas using electromagnetic radiation, and the extraction proceeds in an electrolytic cell.

[0021] The reader will appreciate from FIGS. 3-5 and the description below that the presently disclosed invention addresses many of the shortcomings of conventional processes. For example, FIG. 3 describes the general operation of the invention. The invention, 300, comprises a reactor 301. An electrolytic cell within the reactor comprises an anode, 302, and a cathode, 303. The reactor is charged with a noble gas and the feedstock, 308. The noble gas is then ionized by exposing the gas to electromagnetic radiation. The ionized gas then functions as an electrolyte, 305. Upon ionization, an electrostatic potential, 309, is applied across the anode, 302, and the cathode, 303. Positive ions from the dissociated feedstock, 308, are reduced at the cathode, 303, and negative ions are oxidized at the anode, 302. The oxidized substance exits through the top of the reactor 301.

The embodiment shown in FIG. 3 is a continuous process where a conveyor (not shown) acts as the cathode, and the desired metal is collected on the conveyor.

[0022] FIG. 4 depicts another embodiment of the invention that includes the process and elements above, where the cathode of the electrolytic cell is a wire mesh 403, and the electromagnetic radiation is generated by a magnetron 401. The electromagnetic radiation is directed to the reactor, 301, via a waveguide, 404. In this embodiment, the gas and feedstock, 308 (not shown), are continuously fed into the reactor, 301, and the gas is ionized into an electrolyte, 305, via the electromagnetic energy generated by the magnetron 401. The gas/electrolyte 308 acts as a carrier for the feedstock. The desired metal is collected on the wire mesh cathode. The other materials exit through the top of the reactor, 301.

[0023] FIG. 5 depicts the preferred embodiment and best mode to practice the invention. This embodiment uses a fluidized bed, 504. A process gas, 502, is continuously fed into the inlet, 402, along with a metal oxide or metal sulfide feedstock 503. The gas acts as a carrier medium. The reactor is continuously exposed to electromagnetic radiation generated by a magnetron, 501; via waveguide, 404, the gas is ionized, creating the electrolyte, 305. Where the electrolyte 305 acts as a carrier medium. The 308 feedstock is fluidized and suspended within the ionized gas, leading to dissociation. The dissociation occurs in an electrostatic field produced between the anode 302 and cathode 403. Positive ions from the dissociated feedstock, 308, are reduced at the cathode 403, and negative ions are oxidized at the anode 302. The oxidized substance exits reactor 301 through the top of the reactor 301 and 505.

[0024] The disclosure above encompasses multiple distinct inventions with independent utility. While each of these inventions has been disclosed in a particular form, the specific embodiments disclosed and illustrated above are not to be considered in a limiting sense as numerous variations are possible. The subject matter of the inventions includes all novel and non-obvious combinations and sub-combinations of the various elements, features, functions, and/or properties disclosed above and inherent to those skilled in the art pertaining to such inventions. Where the disclosure or subsequently filed claims recite “a” element, “a first” element, or any such equivalent term, the disclosure or claims should be understood to incorporate one or more such elements, neither requiring nor excluding two or more such elements.

Definitions

[0025] The following definitions apply herein unless otherwise indicated.

[0026] “Comprising,” “including,” and “having” (and conjugations thereof) are used interchangeably to mean including but not necessarily limited to and are open-ended terms not intended to exclude additional elements or method steps not expressly recited.

[0027] “Coupled” means connected, either permanently or releasably, whether directly or indirectly through intervening components.

[0028] “Communicatively coupled” means that transference is enabled between the coupled devices. Examples of transference include the direction of electromagnetic radiation to a specific area or mechanical energy, pressure, or heat directed or applied to a particular area.

[0029] “Electroextraction” means metals are extracted from a substance using an electric current.

[0030] “Electro-deoxidation” means oxidants (e.g., Oxygen, Sulfur) are removed from a substance using an electric current.

[0031] “Plasma-deoxidation” means oxidants (e.g., Oxygen, Sulfur) are removed from a substance using a metastable compound created in a plasma.

[0032] “Metastable compound” is a chemical compound formed under specific conditions by one or more atoms that are in an excited, temporarily stable state.

[0033] Applicant(s) reserves the right to submit claims directed to combinations and sub-combinations of the disclosed inventions that are believed to be novel and non-obvious. Inventions embodied in other combinations and sub-combinations of features, functions, elements, and/or properties may be claimed through amendment of those claims or presentation of new claims in the present application or in a related application. Such amended or new claims, whether they are directed to the same invention or a different invention and whether they are different, broader, narrower, or equal in scope to the original claims, are to be considered within the subject matter of the inventions described herein.

1. A device for electrolysis, comprising: a reactor, an electrochemical cell, a means for generating electromagnetic radiation, and a waveguide; where the electrochemical cell is disposed within the reactor, where the means for generating electromagnetic radiation is coupled to the waveguide and the waveguide is communicatively coupled to the reactor.

2. The electrolysis device of claim 1, wherein the electrochemical cell further comprises an anode, a cathode, an electrolyte, and a means to apply an electrical potential across the anode and cathode, where the anode and cathode are separated from each other and immersed in the electrolyte.

3. The electrolysis device of claim 2, wherein the electrolyte comprises an ionized noble gas.

4. The electrolysis device of claim 3, wherein the ionized noble gas is an ionized species from the group consisting of Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), or Radon (Rn).

5. The electrolysis device of claim 4, wherein the cathode is a conveyor belt.

6. The electrolysis device of claim 4, wherein the cathode is a flat plate.

7. The electrolysis device of claim 4, wherein the cathode is a feedstock confined by a conductive mesh.

8. An electroextraction process comprising:

- a. providing a feedstock;
- b. charging the feedstock and a noble gas into the electrolysis device of claim 4;
- c. creating an electrolyte by ionizing the noble gas using electromagnetic radiation;
- d. dissociating and separating the feedstock by applying an electric potential across the anode and cathode;

e. collecting the desired metal at the cathode;

f. discharging the non-metal and gas from the electrolysis device.

9. The process of claim 8, wherein the feedstock comprises a metal oxide or metal sulfide.

10. The process of claim 8, wherein the noble gas is selected from the group consisting of Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), or Radon (Rn).

11. The process of claim 8, wherein the process is continuously collecting metal on a cathodic conveyor belt.

12. The process of claim 8, wherein electrolyte and feedstock are held in a fluidized bed between the anode and cathode.

13. An electro-deoxidation process comprising:

- a. providing a feedstock;
- b. charging the feedstock and a noble gas into the electrolysis device of claim 4;
- c. creating an electrolyte by ionizing the noble gas using electromagnetic radiation;
- d. dissociating and separating the feedstock by applying an electric potential across the anode and cathode;
- e. reducing the desired metal in place at the cathode;
- f. discharging the non-metal and gas from the electrolysis device.

14. The process of claim 13, wherein the feedstock comprises a metal oxide or metal sulfide.

15. The process of claim 13, wherein the noble gas selected from the group consisting of Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), or Radon (Rn).

16. The process of claim 13, wherein electrolyte and feedstock are held in a fluidized bed between the anode and cathode.

17. A process for plasma-deoxidation, comprising:

- a. a reactor;
- b. providing a feedstock;
- c. charging the feedstock and a noble gas into the reactor;
- d. ionizing a noble gas using electromagnetic radiation;
- e. creating a metastable noble gas compound by fluidizing the feedstock with the ionized noble gas;
- f. retaining a metal agglomeration in the reactor;
- g. discharging the metastable noble gas compound from the reactor.

18. The plasma-deoxidation process of claim 17, wherein the noble gas compound comprises a non-metal element and a noble gas.

19. The plasma-deoxidation process of claim 17, wherein the noble gas is an ionized species from the group consisting of Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), or Radon (Rn).

20. The plasma-deoxidation process of claim 18, wherein the non-metal element is selected from the group consisting of Oxygen (O), Sulfur(S), Fluorine (F), Chlorine (Cl), Carbon (C), Nitrogen (N), Phosphorus (P), Bromine (Br), or Iodine (I).

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